SAMPLE

History of Astronomy: Golden Age of Planetary Exploration

Voyager to Cassini

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Free sample: Introduction & Chapter 1

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Introduction

For nearly all of recorded history, the planets were points of light. A Babylonian priest, a Greek philosopher, a Renaissance chart maker, and a Victorian gentleman with a fine telescope all looked at the same five wandering stars, and all of them saw the same thing: a bright dot that moved against the fixed background. Telescopes enlarged the dots and revealed markings, polar caps, cloud belts, and hazy circles of light that different eyes drew differently.

What if one morning, the points of light began to speak?

The transformation came so fast that a single human lifetime held it all. A child born in the 1920s grew up in a world where Mars was a telescopic sketch argued over by amateurs. By the time that child reached middle age, machines built by human hands had flown past Mars and photographed its craters. By retirement age, machines had landed there. A person could read about Percival Lowell's canals as a schoolchild, watch the Apollo astronauts walk on the Moon in midlife, and see the Voyager photographs of Neptune's deep blue disk as a pensioner. No generation before or since has watched the sky change so completely.

Astronomers call the period roughly from 1962 to 1989 the golden age of planetary exploration, and the name is honest. In those twenty-seven years, spacecraft visited every planet known to the ancients. Venus, the Evening Star, turned out to be a crushed, corrosive inferno. Mars surrendered its dry, ancient riverbeds. Jupiter revealed a churning ice-moon ocean and a volcano-covered satellite. Saturn's rings resolved from smudged streaks into a million braided spirals of ice. Uranus was found lying on its side, rolling around the Sun like a barrel. Neptune, the last of the classical planets, finally showed its face in 1989, wearing a storm as wide as the Earth. The Moon, too, belongs to this story: the first world humans ever explored in person, and the proving ground for everything that followed.

This book tells the story of that age through its machines and its people. You will meet the stubborn engineers who rebuilt planetary navigation from scratch, the mission controllers who worked years-long shifts measured in the positions of planets rather than clock hours, and the scientists who placed bets on what the cameras would find and lost most of them. You will see how a failed Venus probe of 1962 differs from a triumph only by the tightness of a few screws, and how the grandest tour of the outer solar system was scheduled not by ambition but by a rare alignment of planets that would not return for another 176 years.

The story is told for curious readers of every kind. No mathematics is needed:

every technical idea gets a plain-language explanation the first time it shows up. What is assumed is only the oldest of human pleasures: the desire to know what lies beyond the next hill. The hills in this story are hundreds of millions of kilometers tall, and the travelers are made of aluminum, wire, and patience.

A warning before we set out. The golden age was also an age of hard failures, and this book does not hide them. Rockets exploded on the pad, probes fell silent in deep space, and some worlds answered our questions with silence or with answers nobody wanted. The machines that succeeded were the survivors of a long and expensive education. That education, paid for with failures as much as triumphs, is part of the legacy of the age, and passing it on is this book\u2019s main purpose.

A note on the name. Calling 1962 to 1989 a golden age is a judgment made afterward, the way historians speak of a golden age of sail or of radio. The people living through those years did not experience them as a single arc; they experienced deadlines, failures, budgets, and arguments. What hindsight sees is the density of firsts: in no other era did humanity gain so many new worlds in so short a time. Some writers prefer a longer golden age that includes the Moon race at the start and the missions of the 1990s at the end; this book uses the span from the first successful interplanetary probe to the last first-visit of a classical planet, with the crewed lunar programme as its opening chapter of context. Boundaries are instruments, not walls, and the age's ideas flow across them freely.

One more scope note before we begin. This book is about the exploration of the planets by machines, and it spends its pages where the age spent its energy: Venus, Mars, Mercury, the giants, and their moons. Comets and asteroids, the Sun, and the outer darkness beyond Neptune each have their own exploration stories, and where they intersect the planetary programme, they appear; where they do not, they wait for other books. The aim here is depth on one theme rather than a checklist of every flight.

Chapter 1: The Discovery of Golden Age of Planetary Exploration

Every age of exploration needs a doorway, and the doorway to the planets opened on an October night in 1957, over a launch pad in the Kazakh steppe. A polished metal sphere, barely wider than a beach ball, rose into the dark on a column of fire, circled the Earth in ninety-six minutes, and announced itself to the world with a steady radio beep. Sputnik 1 carried no cameras and no instruments beyond a temperature sensor and a transmitter. It did not need them. Its mere existence changed everything, because it proved that a machine made by human hands could be placed above the atmosphere and made to stay there.

The radio beeps were heard, with varying emotions, around the planet. To the engineers who built it, Sputnik was a triumph of the R-7 rocket, designed to carry warheads and proving equally capable of carrying satellites. To the politicians of the United States, the beeps were a shock, a demonstration that an adversary could deliver a payload anywhere on Earth. To the scientists of both nations, they were something else entirely: an invitation. If a machine could orbit the Earth, a bigger machine could escape the Earth entirely. And beyond the Earth, waiting in plain view, were the planets, five of them, circling the same Sun, close enough to reach and yet utterly unexplored.

The first target chosen was the nearest: the Moon. It was near, it was a superb military-civilian prestige target, and the physics of reaching it, while demanding, were understood. Between 1959 and 1976, the Soviet Union's Luna programme chalked up a string of firsts that reads like a fever dream of the Space Age. Luna 1, launched in January 1959, missed the Moon by several thousand kilometers and became the first craft to escape Earth's gravity and orbit the Sun. Luna 2, eight months later, hit the Moon squarely and scattered the Soviet pennants it carried across the lunar surface at the moment of impact. Luna 3, in October of that year, swung around the far side of the Moon and photographed it, revealing for the first time the hidden hemisphere, which turned out to be almost empty of the dark plains that mark the face we see.

The crewed race took the headlines, and the landing of Apollo 11 in July 1969 stands rightly among the signature achievements of the century. But alongside the crews of astronauts stood a quieter army of machines, and while the crews could only reach the Moon, the machines had farther destinations in mind. In 1961, even before a human had orbited the Earth, planners at the Jet Propulsion Laboratory in Pasadena, California, had proposed a spacecraft that could fly past Venus, the closest planet. The proposal became Mariner 2, and its story is the true beginning of the golden age.

The road there was anything but smooth. The first American attempts at Venus and Mars failed in ways that would become grimly familiar to early planetary engineers. Mariner 1, launched in July 1962, lasted 293 seconds before a guidance error forced its destruction; the error became legend, blamed by newspaper wags on a dropped hyphen in the guidance equations, though the truth involved an ambiguous transcription in the coded instructions. Its twin, Mariner 2, was launched a month later on an overcast August morning and became the first fully successful interplanetary probe in history. On December 14, 1962, after a cruise of more than a hundred days across some 290 million kilometers of space, it swept past Venus at 34,000 kilometers and, in a brief encounter measured in minutes, rewrote everything humanity knew about the cloud-covered planet.

The instruments of Mariner 2 were primitive by later standards: microwave and infrared radiometers, a magnetometer, particle sensors. But they answered the first and most urgent question about Venus: is it a warm, damp twin of Earth, as many mid-century astronomers hoped? The radiometers said no, emphatically. The surface beneath the clouds measured over 400 degrees Celsius, hot enough to melt lead. Venus, the brightest and most beautiful star in the human sky, the planet that had inspired more wishful thinking than any other, was a furnace. The disappointment was real, and so was the lesson that would govern every later mission: the planets do not care what we hope. They are what they are, and the only way to find out what they are is to go and measure.

The Soviets, meanwhile, pressed a parallel campaign toward Venus that would ultimately succeed spectacularly. Venera 1, launched in 1961, went silent before arrival. But the programme learned. In 1966, Venera 3 became the first human object to reach the surface of another planet, striking Venus though its radio had already died. Venera 4, in 1967, became the first craft to enter another planet's atmosphere and return data during the descent, transmitting until the crushing pressure silenced it high above the ground. The measurements revealed an atmosphere overwhelmingly made of carbon dioxide, with surface pressures that the engineers had not believed possible. Piece by piece, the furnace was being mapped, and the measurement of its hostility was itself a discovery: the second planet from the Sun was not a second Earth but a warning of what a runaway greenhouse could do.

By the mid-1960s, then, the pattern of the age was set. Launch windows to the planets opened on schedules fixed by orbital mechanics, and each window was met with a volley of probes from two competing nations. Most failed; some succeeded; every success rewrote a chapter of astronomy that had stood for centuries. The doorway

opened by Sputnik had become a highway, and the highway pointed at every planet in the sky.

The crewed lunar programme deserves its paragraph here, both because it shared the decade and because it seeded the planetary missions with people and techniques. President Kennedy's 1961 pledge to land a person on the Moon before the decade's end assembled the largest engineering workforce ever devoted to a peaceful goal, and its descendant technologies, from the Saturn family of boosters to the deep space tracking network, served the planetary programme in turn. The Rangers and Surveyors that photographed and soft-landed on the Moon in the mid-1960s rehearsed the steering, targeting, and landing methods the planetary probes would need.

And one moment in that programme belongs to planetary science in a deeper sense than its planners intended: in December 1968, the crew of Apollo 8 became the first humans to see the whole Earth at once, hanging above the lunar horizon, and the photograph they brought home did more to establish the Earth as one planet among the family than any instrument's data. The exploration of other worlds, it turned out, was also a way of seeing our own.

The far side of the Moon also figures in the story of methods. Luna 3's crude photographs of 1959, developed aboard the craft in an automated darkroom, transmitted by mechanical scanning, showed a hemisphere almost innocent of the dark plains that dominate the familiar face. The result settled an argument about the Moon's two-faced asymmetry that telescopes had never resolved, and it demonstrated the pattern that would define the age: a machine, sent where no eye can go, changes the question on the ground. When the later Luna and Apollo landings sampled the Moon's rock directly, the pattern matured into its final form, measurement first, speculation after, and the whole planetary programme inherited it.

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